Once Bitten, Twice Shy? The Lasting Impact of Enforcement on Tax Compliance

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Abstract

We examine the impact of enforcement on subsequent compliance behavior by taxpayers. Exploiting waves of randomized audits by the Internal Revenue Service from 2006 to 2009, we find three long-run responses by taxpayers. First, audits increase tax payments substantially in following years, but this effect is short-lived when third-party reporting is not available. Second, taxpayers with high income volatility revert to their preaudit behavior quickly. Third, sophisticated taxpayers are affected less by enforcement. These responses reveal how taxpayers perceive the enforcement risk and change their noncompliance according to the dynamics of the information barrier between them and the enforcement agency.

1. Introduction

Tax noncompliance represents a significant legal issue. In the United States, taxpayers voluntarily pay only about 81.7 percent of their tax liability each year according the most recent estimate by the Internal Revenue Service (IRS). Tax enforcement, mostly via audits, helps to recover an additional 2 percentage points of tax liability. The remaining 16.3 percent, or roughly $406 billion each year, never reaches the federal budget because of tax noncompliance (IRS 2016). While we know that enforcement directly improves tax collection by 2 percentage points, we have very little knowledge about the indirect deterrence effects that contribute to the 81.7 percent that is paid voluntarily.

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Understanding deterrence effects has been a central question in the field of law and economics, particularly after the seminal article by Gary Becker (Becker 1968). It has been a consensus that punishment risk provides the foundation for deterrence effects (Pogarsky and Piquero 2003). However, there is no consensus on how enforcement deters subsequent crimes. For example, Gibbs (1975) and Tittle (1980) and ensuing studies argue that punishment discourages future illegal activities. Yet Sherman (1993), Paternoster and Piquero (1995), and Piquero and Pogarsky (2002) suggest that criminals are more likely to commit crimes again immediately following punishment.

Our understanding of why people pay or do not pay tax has grown over time. Starting from early conceptual frameworks (for example, Allingham and Sandmo 1972), we have gained valuable insights from lab experiments (for example, Alm and McKee 2004; Fortin, Lacroix, and Villeval 2007; Choo, Fonseca, and Myles 2014). Most recently, a new strand of research using field data and experiments highlights the importance of third-party information in ensuring that individuals pay tax (Kleven et al. 2011; Carrillo, Pomeranz, and Singhal 2017; Pomeranz 2015; Slemrod et al. 2017; for recent surveys, see Alm 2012; Slemrod 2016).

Tax compliance provides a useful setting for us to examine the deterrence effects of enforcement. Alm, Jackson, and McKee (2009) note that audits can impact taxpayers in three ways: a direct effect (that is, adjustments on the audited return), a direct (specific) deterrence effect (that is, taxpayers who are audited change their subsequent behavior), and an indirect (general) deterrence effect (that is, effects on those who are not audited).

Our focus is on the direct deterrence effects of audits, that is, the impact of audits on subsequent tax payments by the audited individuals. Three features of IRS policy and our data make this possible. First, the IRS conducts intermittent random audits and keeps systematic records of them. Second, the IRS provides accurate data on subsequent tax payments in each year following the audit, even when there is no subsequent audit. Third, these data constitute a panel of the entire population of individual taxpayers over time, which allows for rigorous empirical analysis.

Conceptually, the long-term response of taxpayers to being audited has not been settled. In theory, experiencing an audit would lead taxpayers to update their perceived probability of audit in a Bayesian manner and thus pay more tax (for example, Dubin, Graetz, and Wilde 1990). However, Maciejovsky, Kirchler, and Schwarzenberger (2007) and Kastlunger et al. (2009) argue for the opposite: a taxpayer would pay less tax right after an audit because she may believe that the auditor is unlikely to target her again soon.

Empirically, three new studies on long-term tax responses using tax data also show very different results. DeBacker et al. (2015) find that US firms reduce tax payments immediately after being audited and then increase them gradually to preaudit levels. In contrast, Advani, Elming, and Shaw (2015) find that self-assessed taxpayers in the United Kingdom increase their tax payments after being audited, while National Taxpayer Advocate (2015), a report that uses one wave

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of randomized audits, finds increased reported taxable income among the self-employed who had positive audit adjustments but finds negative impacts among those who had no additional tax assessment from audits.

We use randomized, comprehensive, multiwave administrative audit and tax return data to improve on these studies in several important ways. In DeBacker et al. (2015), a series of control variables are used to account for the fact that operational audit likelihoods are endogenous to the outcomes expected by the tax authority. By relying on randomized audits, the selection into the treatment and control groups is made exogenous, and audited taxpayers can be compared directly with those not selected for audit. While Advani, Elming, and Shaw (2015) do use randomly selected audits, they also have data on only those taxpayers who had to submit a self-assessment of their tax. The reliance on the self-assessment of taxpayers means that the results are not generalizable to the entire taxpaying population; it is generalizable to only a self-selected group that accounts for about a quarter of the taxpaying population and mainly consists of the self-employed, company directors, pensioners, and very high-income taxpayers. They also construct their control group out of individuals who are audited in the future rather than those who were never selected for audit. This requires them to make assumptions about the intertemporal stability of the audit process and raises significant concerns about survivorship bias. In contrast, our use of the universe of all tax returns allows us to construct a control group that covers almost every taxpayer and to follow its members into the future for an arbitrary number of years. Finally, National Taxpayer Advocate (2015) focuses on schedule C filers who were audited in 2007. In addition to expanding the scope of the type of returns audited, we also expand the scope over the number of waves of audits. Because of major macroeconomic events in the late 2000s, it is difficult to say to what extent any economic behaviors are typical. The more waves of audits we are able to incorporate, the more confident we can be that we are observing typical responses.

In this paper, we provide a theoretical framework and empirical results that extend the literature in several ways and that help provide further evidence to answer some of these unresolved questions. To start, we consider the link between third-party information reporting and long-term responses to audit. This link has not been studied, but we believe it is central to understanding the deterrence effects of enforcement against tax evasion and similar illegal behaviors.

In particular, we propose that the difference in information held by the tax authority and taxpayers determines the long-term impact of audits. In practice, the tax authority usually has imperfect information about the true income of a taxpayer. The tax authority makes attempts to update this information from two main sources, namely, third-party reporting and tax audits. Although the information from a third party (for example, an employer) tends to be updated annually, the audit information is updated only when an audit is conducted. When audits are not conducted, information from previous audits gradually becomes outdated.

The speed of updating and outdating of income information allows those who

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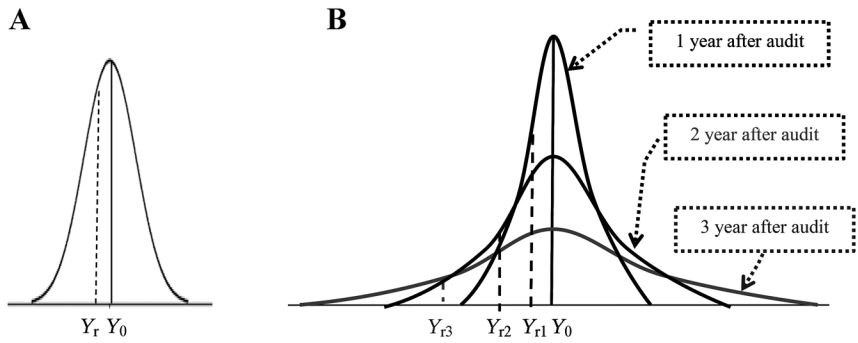


Figure 1. Long-term effect of an audit on underreporting income. *A,* Wage earner; *B,* business owner.

wish to evade tax to form a perceived audit risk and manipulate their income reporting strategically. Figure 1 describes the dynamics of this process. It shows the conditional income distributions of a taxpayer as perceived by the auditor. Figure 1*A* is for an income source that has a low variance; Figure 1*B* is for an income source with a high variance. In Figure 1*B*, the distribution is most highly concentrated in the year after audit, as the information from the audit is still relatively informative about the distribution of true income (Y_0). However, as time passes, the spread of the distribution increases as the information from the audit becomes more outdated. This increase in the spread of the distribution results in more underreported income as the reduction in information to the IRS results in more scope for taxpayer noncompliance. Thus, we expect to see reported income (denoted Y_r) decline as the spread in the conditional income distribution increases in the number of years since audit. For income sources that have less variance or volatility (or that are subject to more third-party reporting), the degree to which audit information becomes outdated is much lower. Thus, we should expect a strong effect of audit on income sources with high variance and little third-party reporting (where the information gleaned from audit is particularly important), but we also expect that this effect declines more quickly than the effect on sources of income with more third-party reporting and less variance or volatility.

We identify three drivers of the probability of audit perceived by taxpayers. First, when third-party reporting is not available, audit information is most useful for the enforcement authority, but it can become outdated quickly. Second, higher income volatility makes this outdating process faster and therefore reduces the deterrence effect of tax audits. Third, more sophisticated taxpayers are able to exploit the difference in information more effectively. Therefore, our predictions are that the effect of audits on tax compliance will be more short-lived and less effective when there is less third-party reporting, when taxpayers have

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higher income volatility, and when taxpayers have higher levels of tax sophistication.

To test for these predictions, we use data from the IRS's National Research Program (NRP). The NRP began conducting random audits of individual tax filers starting in tax year 2001 and implemented annual random audits starting in 2006. To these data, we merge returns from the universe of filers from 2000 to 2012, which allows us to examine the impact of audits on individual taxpaying behavior for a period of up to 6 years after an audit.[1] For the first time, data allow us to study the long-term responses to audit among both wage earners and filers who are self-employed and measure the responses to audit across a large set of income and deduction items. This is key to understanding the interaction between third-party information and audits in the long run. The treatment, an IRS audit, is randomized through the NRP; thus, our empirical strategy is straightforward. Using the sampling weights that the IRS uses to select individuals for an NRP audit, we construct a nationally representative sample of audited individuals. We pair this representative sample with a random sample of individuals drawn from the same population of tax filers. We then compare the tax filings of these two groups before and after the audit year. This exercise allows us to study the impact of three drivers of long-term tax behavior: third-party information, income volatility, and taxpayer sophistication. We summarize the key results for these drivers below.

Third-Party Information. We find that an audit increases subsequent reported wage income by 1.3 percent and sole-proprietorship income, reported on schedule C of form 1040, by 14.2 percent on average. Thus, the effect of audit is much larger when there is less third-party reporting, which is consistent with Kleven et al. (2011). Importantly, we show that audits have a long-term effect on tax reporting. An audit increases reported taxable income by an average of $1,185 per year, equivalent to about 2.9 percent of the average taxable income. Thus, the direct revenue gain from the audit adjustments understates the true revenue gain. Accounting for changes in tax reporting in the 5 years following an audit increases the revenue gain by more than 100 percent.

An interesting and novel finding is that there are differential effects on the persistence of the effects of audits over time. The IRS data show that, after an audit, reported schedule C income increases sharply by more than 16 percent but then falls quickly in subsequent years. The effect of audit on reported schedule C income in the fifth year after an audit is only about one-third of the effect in the first year after an audit. In contrast, reported wage income increases by only about 1 percent in the first year following audit but continues to increase in the following 2 years to approximately 2 percent, after which it remains relatively constant.

Income Volatility. Beyond third-party information, we propose income vol-

[1] The Internal Revenue Service (IRS) defines an audit as "a review/examination of an organization's or individual's accounts and financial information to ensure information is being reported correctly according to the tax laws and to verify the reported amount of tax is correct" (IRS, IRS Audits, [https://www.irs.gov/businesses/small-businesses-self-employed/irs-audits]).

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atility as an important factor enabling tax evasion. To illustrate, consider that when the tax authority finds it harder to predict the income of a plumber than the salary of a teacher, it is not only because the plumber may have less income subject to third-party information reporting but also because the plumber's income may be more volatile year over year than that of a teacher. If the plumber reports a large change in income, this volatility can reduce the auditor's certainty as to whether the change represents tax avoidance or real economic activity. A higher degree of income volatility increases the rate at which the information gleaned from an audit becomes outdated. Thus, two factors, income volatility and third-party information, may correlate and confound previous studies that focus on the role of information reporting. We include measures of income volatility in our analysis and find that volatility mitigates the persistence of the effects of audit on reported income. This empirical result holds even when we control for third-party information reporting. The implication is that the effects of audits on tax-paying behavior diminish more quickly for filers with higher income volatility, holding constant the level of third-party information reporting across those filers.

Taxpayer Sophistication. Furthermore, we find that variation in the effect of audit across filers is correlated with measures of taxpayer sophistication. Tax literacy, familiarity with audits, and the use of tax preparers reduce the effectiveness of tax audits. These results show that those taxpayers with a better understanding of the tax code respond more strategically to their audit experience.

The results above contribute to an important literature on tax enforcement started by Allingham and Sandmo (1972). Several early studies consider the impact of audit rates on individual taxpayers' compliance. These include Tauchen, Witte, and Beron (1993), Dubin, Graetz, and Wilde (1990), and Witte and Woodbury (1985), all of whom find that increases in audit rates increase compliance. Marginal tax rates may also impact evasion since they affect the value of misreported income. Evidence of this effect has been provided by Clotfelter (1983) and Feinstein (1991).

Our analysis is built on several important works on the role of third-party information. Feldman and Slemrod (2007) find that documented income is much less likely to be evaded. Kleven et al. (2011) point out the importance of third-party information, which makes Danish taxpayers unable, rather than unwilling, to cheat. Pomeranz (2015) and Slemrod et al. (2017) show that third-party information reduces tax evasion among Chilean and US businesses. Our results support these previous findings, but we also show that the large impact of an audit on reported income for people without third-party reporting found in previous studies diminishes very quickly over time.

Our results also contribute to work that examines the indirect deterrence effect of enforcement against tax evasion and other crimes. These studies include Alm and Yunus (2009), which finds a role for norms and learning in tax evasion in the United States, and Dubin (2007), which calculates the deterrence effect of audits. We show that this deterrence effect is mitigated by taxpayers' income volatility and sophistication.

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The paper is organized as follows. Section 2 describes the conceptual framework. Section 3 describes the NRP, and Section 4 describes the data. Section 5 presents information on tax compliance in our data. Section 6 presents our main empirical results, and Section 7 reports results for particular income and deduction items and the relationship between tax compliance and income volatility. In Section 8, we offer evidence that the response to audit is affected to a great extent by the filer's level of tax sophistication. Section 9 concludes.

2. Conceptual Framework

To guide our empirical analysis of the tax compliance behavior of taxpayers following audit, consider the model of tax evasion introduced by Allingham and Sandmo (1972). The Online Appendix presents an extension of this model in which beliefs about the probabilities of audit evolve depending on one's interaction with the tax system, including the amount of interaction with the tax authority and audit experiences.[2] Such a model yields a number of interesting comparative statics. As in Allingham and Sandmo (1972), reported income is always less than or equal to true income, reported income is increasing in the penalty rate, and the effect of the marginal tax rate on reported income is indeterminate unless more assumptions are made about the utility function of the taxpayer.[3] In addition, and of more direct interest to our study, are the following proposition and corollaries. (The proof of proposition 1 and these corollaries are in Online Appendix Section OA1.)

Proposition 1. Reported income is increasing in the perceived probability of audit.

Corollary 1. The effect of an audit on reported income is ambiguous.

Corollary 2. Reported income is higher for third-party reported income.

Corollary 3. Reported income decreases in the variance of the taxpayer's income.

Corollary 4. The change in the taxpayer's perceived probability of audit is decreasing in the number of tax filings she has experienced.

Our empirical approach is centered around testing these relationships, in particular how the audit experience affects reported income and how this varies across income sources and filer types. Given the contrasting evidence from the field and laboratory, corollary 1 leads to an ambiguous result. Our study provides evidence on this empirical question. Corollary 2 suggests that filers with more third-party reported income are more compliant. We look at evidence of this but also consider the differential impacts of audits across different sources of income.

[2] A formal treatment of this framework is provided in Section OA1 of the Online Appendix.

[3] One can see this from the first-order condition. A higher marginal tax rate increases the benefits of evasions. However, since the penalty is proportional to the tax liability avoided, the expected costs to evading another dollar are also increasing in the marginal tax rate.

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If audits induce compliance, we would expect the responses to be larger from taxpayers or income sources that were less complaint prior to audit. That is, we would expect larger changes in reported income after audit from income sources with less third-party reporting.

In the context of the lasting effects of audit, we consider what corollary 3 implies about the responses to audit from those with more highly variable income sources vis-à-vis those with sources of income that are less variable. Finally, corollary 4 suggests that we should find a smaller effect of audit among groups of filers who are more familiar with the tax system: those with tax sophistication. Our test of corollary 4 is thus to test whether groups with more experience filing—that is, more tax-literate groups—respond less to the audit event than do other groups.

To summarize, our theoretical model implies four results that we can examine in the data. First, the direction of the effect of an audit on subsequent income reporting is ambiguous. Second, to the extent that income reporting responds to audit, we observe more of a response for taxpayers with income sources that have less third-party reporting. Third, the responses to audit will be stronger and more transitory for taxpayers with sources of income with less third-party reporting and more variance or volatility. Fourth, the effect of audit will be smaller among groups with more tax sophistication.

3. The National Research Program

The NRP conducts audits on a stratified, random sample of the filing population in order to inform IRS enforcement policy. In this section, we discuss the similarities and differences between NRP and operational audits so that the reader can judge the external validity of our research using NRP audits.

The NRP examination guidelines (IRS 2012a) state that NRP audits should be conducted in a manner similar to that for operational audits. The guidelines specify this similarity in many aspects, including depth, required filing checks, amount of data collected, judgment standards, referrals to other IRS units, the examiner's responsiveness, and availability to meet with the taxpayer.

The NRP audits are also similar to operational audits in terms of how the examination location is determined. Most audits are conducted in person by an auditor, while some are performed via correspondence, and some returns are simply accepted as is without further contact with the taxpayer. Note that in the third style of audit, the taxpayer is generally unaware of the audit, and it presumably has no behavioral effect.[4] All three methods are used in both operational and NRP audits.

[4] This third type of audit accounts for approximately 5 percent of the audited sample. We note that the National Research Program (NRP) randomly chooses tax returns to audit but nonrandomly determines the examination method. Therefore, we include this last type in the sample to ensure the randomness of the audit treatment. This means our estimates are the effect of the intent to treat with an audit rather than the effect of the treatment on the treated.

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The NRP audits are, however, different from operational audits in several ways. Unlike the operational audits, the NRP audits are conducted to collect detailed compliance data on all the potential issues on the return. As a result, they may cover parts of the tax return for which there is no suspected noncompliance. Operational audits are focused more narrowly on areas of suspected noncompliance (Black et al. 2012). Interestingly, taxpayers report higher satisfaction from NRP audits than from operational audits (Brown and Johns 2007).

The most important distinction between NRP audits and operational audits, which is relevant for the external validity of our study, is that the taxpayer is informed that an NRP audit is random and therefore not related to any action taken by the taxpayer. Furthermore, the infrastructure of the NRP system is set up for measurement and statistical analysis, not for building profiles of individual taxpayers (IRS 2017; I.R.S. IRM 4.22.1 [September 6, 2017]). While the understanding of what this means (and thus the behavioral response) varies between taxpayers, we would expect those taxpayers with greater tax sophistication to be better able to take advantage of these features.

However, the audited taxpayer may learn about the stringency of an audit or the ability of the auditor to identify problems in the tax return. Furthermore, the taxpayer may learn about the tax code and how to file taxes properly after an audit. We also refer to work such as Manoli and Turner (2014), which finds that contact from the IRS is perhaps more important in changing behavior than the content of that contact. While it is likely that an NRP audit will educate the taxpayer about a wider range of such topics, it is difficult to say whether NRP or operational audits educate the taxpayer on the most salient tax topics (in particular, those for which the taxpayer is currently noncompliant or is considering future noncompliance).

In sum, we speculate that after an NRP audit, taxpayers will update their perceptions about both the probabilities and stringency of audit. Taxpayers seem likely to gain information about stringency from both NRP and operational audits, though taxpayers may learn less information about a broader swath of topics under an NRP audit. In terms of the probability of audit, taxpayers seem likely to learn less about their risk of audit from an NRP audit than an operational audit, primarily because undergoing an operational audit is prima facie evidence to the taxpayer that the IRS perceives them to be noncompliant in some way. However, those with less tax sophistication seem more likely to conflate the two forms of audits and respond more strongly to the NRP audits than more sophisticated taxpayers who fully comprehend that NRP audits are random.

4. Data

Our data come from three sources. We discuss each data source in turn and then describe the process by which we merge the data and create our final sample.

First, we use data on audits from the IRS's NRP. In particular, we use the taxpayers' information generated by the audits conducted as part of the NRP's 2006,

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2008, and 2009 waves.[5] The information includes taxpayers' identifiers (the social security number [SSN] of the primary filer), year of the audited return, and the resulting adjustment to selected lines on form 1040.[6] Each of the 2006–9 waves has approximately 15,000 observations.

Second, we use data from the IRS's population of individual income tax returns. These data include many items from the filer's form 1040 and the associated forms and schedules, including all items on the front page of form 1040 and the main line items from most associated schedules. We use these data from the years 2000 to 2012.

Finally, we use data from the IRS's Audit Information Management System (AIMS). The AIMS data contain detailed information on all IRS audits (including NRP and non-NRP audits) from 1996 to the present. We use these data to augment the audit data from the NRP. In particular, the AIMS data allow us to observe variables such as the date the audit began and ended, the hours of the examiner's time put toward the audit, and examiner's characteristics.

We construct our sample with a control group and a treatment group. We create the control group by randomly selecting a .1 percent sample of filers, choosing a different set of 10 four-digit SSN endings for each year (2006, 2008, and 2009).[7] For each of these years, we then select all primary filers who had one of these 10 four-digit endings from the universe of returns filed that year. We create our treatment group by finding the SSN of all primary filers in NRP waves from 2006, 2008, and 2009. Finally, we select all tax returns from the 2000–2012 period for filers in either our sample group or control group.[8] Our final panel thus is composed of a control group of randomly selected filers from the years of the NRP waves (followed over time) and a treatment group of randomly audited filers from the NRP waves (who are also followed over time). Creating our control group in this way (by ensuring that those in the control group filed a return in the

[5] The NRP was not conducted in the years 2002–5. We exclude the NRP waves from 2001 and 2007 and those conducted after 2009. Documentation suggests that the sampling frame and intent of the 2001 wave was too different from later waves to treat them as comparable. Furthermore, constructing the control group for the 2007 NRP wave was complicated by the fact that stimulus rebate checks were sent out in 2008 and that to be eligible for a stimulus check one must have filed a year 2007 tax return. The resulting population of filers for tax year 2007 (who filed taxes in early 2008) was much different than in other years and did not closely resemble the weighted NRP sample. In particular, there was an increase in the number of people who typically did not file tax returns. We attempted to address this anomaly by using the methodology of Ramnath and Tong (2014) to identify those who filed only to claim the stimulus check, but the random sample was still sufficiently different than the NRP sample. The NRP waves later than 2009 have been excluded to ensure that open audits were able to be completed before we drew our data.

[6] Unfortunately, information on penalties and interest owed (if applicable) is not contained in these data.

[7] The sample size is dictated by computational constraints.

[8] While we use preaudit values in all years for consistency, we also estimated equations (2) and (3) using the postaudit values for the year they are available. The results are similar, except that the estimated coefficients are all approximately $2,500 lower. This is consistent with Table 2, where the unconditional audit adjustment is $2,419. In other words, taxpayers report larger tax payments after audit but not as large as the discrepancy found by the auditor.

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Table 1
Observations in the Samples

	N
NRP sample:	
2000–2012	4,364,655,406
2006, 2008, and 2009	404,252,738
Positive adjustment to tax liability	164,338,287
No adjustment to tax liability	210,135,968
Negative adjustment to tax liability	29,778,483
Random sample:	
2000–2012	4,335,270,000
2006, 2008, and 2009	405,187,000

Note. The sample from the National Research Program (NRP) is the treatment group, which has been audited. The random sample is randomly selected from the universe of tax filers. Our data include 2006, 2008, and 2009 NRP waves. The number of observations in the NRP sample is weighted according to the weights used by the NRP.

year the treatment group was audited) allows us to match attrition rates across treatment and control groups.

By combining the random sampling of the NRP with our method of constructing the control group, we are able to avoid many of the pitfalls that would typically affect a panel study of tax compliance. By ensuring that our control group was selected in waves in the same manner as the treatment group, differential attrition cannot be attributed to the fact that we are combining multiple waves of data. Furthermore, because selection into the NRP is random, any behavioral differences between the control and treatment groups cannot be explained by selection into treatment based on observables. We believe that these two factors rule out the vast majority of purely mechanical explanations for observed outcomes.

Using the SSN of the primary filer, we are able to link returns across the three data sources: the IRS population files, the NRP, and AIMS. Thus, in our final panel, we have detailed information on each tax return filed from 2000 to 2012. For the treatment group, we also have detailed information on the characteristics of the audit and the adjustments to tax returns following audit, though we lack information on audits that are not closed by the time we extracted data from the AIMS database. As a result, information from audits not closed by October 2014 is missing in our sample. However, given that our last NRP wave is from 2009 and that well over 95 percent of audits are closed within 2 years, almost all audits were closed. Table 1 summarizes our sample, noting weighted observations in the base year (that is, the year of the NRP wave) and across 2000–2012.[9] Operational

[9] We use weights for both our randomly sampled control group and the treatment group. We weight the control groups by giving each filer equal weight to sum to the total population of filers in the base year. We weight the treatment groups using the NRP sampling weights. This method gives us a number of weighted observations approximately equal to the population of filers in the base year for the NRP sample. We then apply these weights to the filing units for each year they are in the panel.

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audits are rare in our treatment and control groups. The likelihood of receiving an operational audit in either group is under 1 percent, in line with overall IRS audit rates.

While we do observe the dates an audit was opened and closed, we do not know when the filer was notified of the audit or the results of the audit. Thus, we use as our timing convention the number of years since the audited return was filed. For example, for the 2006 NRP wave, the tax year 2006 return was audited. Thus, we consider the tax year 2007 return as being 1 year since the audited return was filed. We use this convention throughout the paper. As a result, one would not expect sharp increases in reported income for all filers in a given NRP wave in a specific year since audit because the duration of audits and the time when filers were notified vary. However, since the vast majority of audits are closed within 2 years, we do expect the effects of audits to fully materialize 2–3 years after the audited tax year.

Throughout, all monetary variables are deflated to 2005 dollars and are 99 percent winsorized. Winsorization of the data is necessary for addressing outliers.[10] The IRS does not edit most of our data sources, and thus data entry and calculation errors by the filers or the IRS agent entering the data are not uncommon. The effect of noisy data is compounded by the fact that the distribution of income in our population is quite skewed, which makes our estimates susceptible to quirks of sampling among the upper end of the income distribution. While our analysis relies on 99 percent winsorization, we are able to obtain very similar results at both 95 percent and 99.9 percent winsorization and qualitatively similar results when income is winsorized at $10 million. These results are reported in Tables OA1 and OA2 in the Online Appendix.

5. Tax Compliance in the United States

The IRS's tax gap measure summarizes aggregate compliance with the US income tax.[11] During 2008–10, the average tax gap was $458 billion (IRS 2016). This gap represents a compliance rate of 81.7 percent. Noncompliance with the individual income tax code is the largest source of noncompliance, accounting for $319 billion of the total gap.

With regard to noncompliance with the individual income tax, IRS (2016) shows that the lowest compliance rates come from taxpayers with income with less documentation. For example, the underreporting of business income, and in particular income from sole proprietors (as reported on schedule C of form 1040), accounts for about half of the individual income tax gap. Looking across income and deduction items, we see that compliance rates fall as withholding and third-party verification decline. Such a pattern is also documented in Danish data by Kleven et al. (2011).

[10] With 99 percent winsorization, we are truncating the values at their 99th percentile. That is, all values at or above the 99th percentile are assigned the 99th-percentile values.

[11] Note that the NRP data we use play a large role in the IRS's estimation of the individual income tax gap.

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Table 2
Summary of Audit Adjustments

	Preaudit Income	Audit Adjustment	Underreported Income	Overreported Income
Taxable income	40,735	4,031	5,167	−1,427
	(76)	(60)	(83)	(17)
Adjusted gross income	48,417	4,502	5,893	−1,512
	(100)	(43)	(81)	(19)
Deductions	2,420	58	726	−961
	(25)	(47)	(60)	(40)
Wages and salaries	42,657	921	1,233	−148
	(84)	(7)	(77)	(23)
Schedule C income	8,400	7,116	8,483	−1,317
	(16)	(73)	(86)	(14)
Schedule D income	3,066	1,721	3,133	−886
	(17)	(16)	(65)	(35)
Schedule E income	11,309	4,512	6,614	−2,186
	(12)	(48)	(76)	(24)

Note. Values are measures of compliance found in the National Research Program (NRP) data. The first column reports means by income and deduction sources and the percentage with those sources of income. The second column reports the average NRP audit adjustment, conditional on nonzero adjustment, and the percentage who report nonzero values. The remaining columns report the average underreported and overreported incomes, conditional on underreporting or overreporting. The nonzero percentages are in parentheses.

Our NRP data allow us to delve more deeply into the data than the tax gap statistics provided by the IRS. Table 2 documents the measures of compliance found in our NRP data.[12] Underreporting of income results in upward adjustments in income or downward adjustments in deductions, and overreporting of income results in downward adjustments of income or upward adjustments of deductions.[13]

Consistent with the compliance results seen in the IRS tax gap reports, the pattern that emerges is that taxpayers with sources of income with the most documentation show the lowest rates of tax noncompliance. We find that noncompliance rates (measured by the fraction of filers with adjustments) are largest for schedule C (sole proprietorship) income, which has no withholding and little third-party verification. Schedule C income is adjusted for about 73 percent of filers who are audited. The rate of noncompliance is lowest for filers with wage and salary income, which is adjusted for about 6.5 percent of audited filers. Underreported income is more frequent than overreporting of income for all sources and is highest for schedule C income. The average amount of underreported schedule C income is $8,483. This figure compares to a mean of $8,400 for reported schedule C income. Wage income is underreported by $1,233 compared with an average of $42,657. Overall, compliance rates are highest for wage income, which has

[12] Table 2 includes only adjustments found on audit. It does not multiply these amounts to account for noncompliance that was not found in an audit.

[13] Note that the second column is the weighted sum of the third and fourth columns.

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a high withholding rate. Compliance rates are also high for capital income (both capital gains reported on schedule D and capital income reported elsewhere), which has third-party reporting on most items but not withholding.

6. Effects of Audits on Subsequent Reported Income

With an understanding of our data and tax compliance in the United States, we now turn to our research question. The objective of the paper is to understand changes in individuals' taxpaying behavior in response to audit. To ensure the robustness of the results, we use posttreatment-difference, difference-in-differences, and within-filer estimation methods.

6.1. Simple Difference-in-Differences Estimates

The randomized controlled nature of the NRP allows us to consider the effects of audit on taxpaying behavior using either a simple postaudit-difference or difference-in-differences estimator. For outcome variables, we look at the reporting of taxable income, wage income, and schedule C income. These three measures provide useful contrast in terms of the amount of third-party information the IRS has regarding each income source. For example, most wages are subject to withholding, schedule C income has very little documentation, and taxable income is a broad measure of overall income, composed of income with different reporting requirements and determined after deductions are reported. Our difference-in-differences estimator of the effect of audit is thus

$$\text{Effect of Audit} = (\overline{Y}_{T,2} - \overline{Y}_{T,1}) - (\overline{Y}_{C,2} - \overline{Y}_{C,1}), \tag{1}$$

where T denotes the treatment group (that is, the NRP sample) and C denotes the control group. The subscripts 1 and 2 denote the preaudit and postaudit periods, respectively. For each, we consider the mean over a span of 3 years. Thus the $\overline{Y}_{T,2}$ value is calculated as the mean of the income source of interest for the NRP sample over the 3 years after audit, and the $\overline{Y}_{T,1}$ value is calculated as the mean of the income source of interest for the NRP sample over the 3 years prior to audit. The means for the control group are constructed in an analogous way.

Table 3 reports the posttreatment-difference and difference-in-differences results. The (reported) taxable income of the audited group increases by \$1,580.99 between the preaudit and postaudit periods, while that of the control group increases only \$383.04. These figures imply that audits increase taxable income by \$1,197.95 on average.

Table 3 also shows that the effect of audits on reported wage income is \$597 on average. This effect is consistent with the understanding that it is more difficult to misreport income that is also subject to withholding and third-party verification (Kleven et al. 2011), either before or after audit, and so the effect is small in per-

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Table 3
Single and Double Differences between the Audited and Nonaudited

	NRP Sample	Random Sample	Difference
Taxable income:			
Preaudit	41,144.14	40,879.17	264.97
Postaudit	42,725.13	41,262.21	1,462.92
Difference	1,580.99	383.04	1,197.95
Wage income:			
Preaudit	43,951.25	43,787.59	163.66
Postaudit	44,967.15	44,206.48	760.67
Difference	1,015.90	418.89	597.01
Schedule C income:			
Preaudit	8,016.93	8,111.85	−94.92
Postaudit	8,290.09	7,331.30	958.79
Difference	273.16	−780.55	1,053.71

Note. Results are posttreatment and difference-in-differences values (in 2005 $) between the audited and nonaudited groups using the National Research Program (NRP) data and the random sample. The means are calculated over 3 years before and 3 years after audit.

centage terms. Note, however, that this response is large relative to the IRS estimated tax gap for wage income.[14]

The largest effect of audits, in percentage terms, is on schedule C income. Table 3 shows this effect to be $1,053 per year, or about 13 percent of preaudit income for the control group.[15] This large effect supports the perception that it is easier to manipulate schedule C income than wage income and that such income reporting is more responsive after an audit.

6.2. Persistence of Audit Effect

Given these results, a question of further interest is how individuals change tax reporting over time after audit: do they increase reported income permanently, or does the initial effect decline as time passes? Figure 2 plots the differences between the mean reported incomes of the audited and control groups. Reported taxable income increases in the first and second years after an audit and remains elevated even after 6 years. Adjusted gross income (AGI) and wage income follow a pattern similar to that of taxable income.

[14] The IRS estimates an evasion rate of 1 percent for wages and salaries (see IRS 2012b). Taking both estimates together could imply that experiencing an audit eliminates any remaining evasion. Alternatively, they could imply that individuals might respond to an audit by overreporting income to reduce the risk of future audits.

[15] This effect is predominantly driven by schedule C income declining substantially among the control group. This is likely because of the fact that the postaudit period spans 2007–12, which includes the depths of the great recession. This recession had a particularly strong negative impact on the earnings of the self-employed, with business or professional net income reported on individual returns declining by almost 10 percent from 2008 to 2009. See IRS, Table A, All Returns: Selected Income and Tax Items in Current and Constant 1990 Dollars, Tax Years 1990–2015 (https://www.irs.gov/pub/irs-soi/15intaba.xls).

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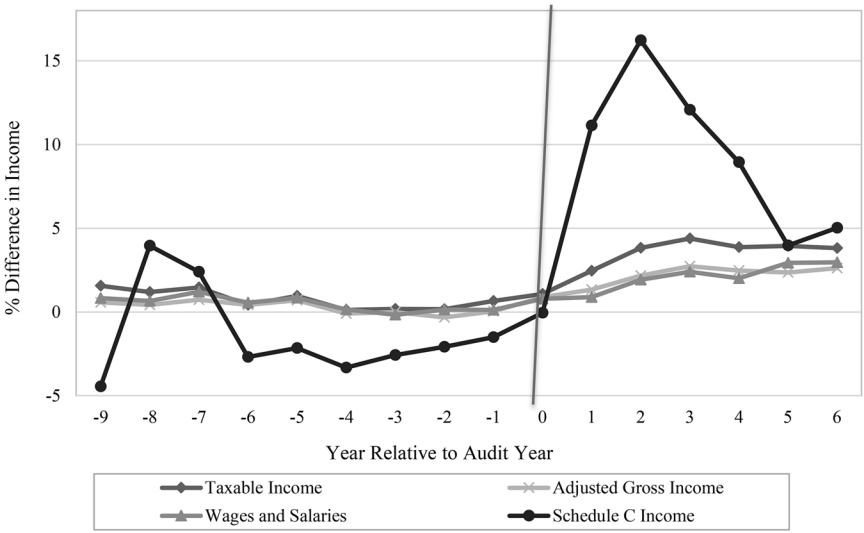


Figure 2. Diffenences in income between treatment and control by income source

As Figure 2 shows, the effect of audits on schedule C income is strong in the first few years after audit. Following the initial upswing in reported schedule C income, it then turns downward toward the preaudit level. This result stands in contrast to the trends in taxable income, AGI, and wage income and suggests further that income with less third-party reporting responds differently to audit.[16] We thus delve more deeply into the varying responses by type of income in Section 7. We next consider models with individual fixed effects, which will allow for identification of the effects of audit from within-filer variation in reported income.

6.3. Within-Filer Estimates

Because we have a panel of tax returns, we can examine changes in individuals' behavior after an audit while controlling for individuals' time-invariant unobserved characteristics. We first estimate an equation of the form

$$\text{Income}_{it} = \beta \text{Postaudit}_{it} + \gamma_i + \eta_t + \varepsilon_{it}, \tag{2}$$

where Income_{it} is a measure of income for individual (taxpayer) i in year t, Postaudit_{it} denotes that the individual was audited during our sample period prior to year t, γ_i is an individual (taxpayer) fixed effect, and η_t is a year fixed effect. In this specification, identification of the effects of audit comes from within-

[16] Note that these postaudit trends are not the result of compositional differences. To show this, we redo this analysis separately for the treatment and control groups from each NRP wave. The figures showing taxable income and schedule C income are available in the Online Appendix and document similar trends across each wave.

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filer changes in reported income between the pre- and postaudit periods, net of trends in income common across the treatment and control groups, which are accounted for by the year fixed effects.

Table 4 reports the results from regressions that estimate the effect of audits on taxable income.[17] The first column shows that audits increase reported taxable income by $1,109, which is statistically significant at the 1 percent level. This result implies that, consistent with the simple difference-in-differences tabulations above, individuals tend to report more income after audit.

The second column examines whether, in our estimation framework, the effects of audit differ with the number of years since the audited tax year by estimating an equation of the form

$$\text{Income}_{it} = \sum_{k=1}^{K} \beta_k (\text{Postaudit}_{it}) \times \left(k\text{Years since Audit}\right) + \gamma_i + \eta_t + \varepsilon_{it}. \quad (3)$$

In this specification, the key explanatory variables are a series of dummy variables that show the difference between the audited and control groups from year 1 through (at most) year 6 after the audited tax year. Table 4 shows that reported taxable income increases quickly during the first 2 years after the audit, with an increase of approximately $1,200, and stays at this level until at least year 6, with all of the effects being statistically significant. Figure 3 summarizes an additional specification in which we include dummies for the 2 years prior to audit. Neither of these preaudit dummies is significant.

Although this tells us about the average response, it does not tell us about how specific taxpayers decide what their response will be. If all taxpayers have the same response, this suggests that there is something in common to all audits that is driving their behavior. More likely, taxpayer response will be affected by which adjustments the auditor makes and how large those adjustments are. While it can be difficult to characterize such a multidimensional interaction as an audit, one obvious comparison we can make is whether the total adjustment to income is upward or downward and how large that adjustment is.

First, we use the information from the audits in a simple way, distinguishing among those who have positive, negative, or no adjustment. We expect that the response to audit would be stronger for filers with a positive adjustment to tax liability than for those with no adjustment. That is indeed the case. We provide evidence for this effect in Figure 4, where we estimate equation (3) on taxable income separately for each of three groups: those with a positive adjustment to tax liability following the audit, those with no adjustment, and those with a negative adjustment.[18] Note that in this specification, we are comparing each outcome with the population as a whole. We see the strongest and most statistically significant response from the group with positive adjustments. Almost all the coefficients for the other two groups are statistically insignificant and have much

[17] While we use preaudit values in all years for consistency, we also estimate equation (2) using the postaudit values for the year they are available. The results are extremely close to those reported here and do not affect our findings.

[18] Regression coefficients are presented in Table OA3 of the Online Appendix.

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Table 4
Effect of Audit on Reported Taxable Income

	Average Effect	Effect over Time
Postaudit	1,109.329**	
	(184.312)	
1 Year since Audit		687.422**
		(186.026)
2 Years since Audit		1,203.155**
		(201.095)
3 Years since Audit		1,424.118**
		(246.485)
4 Years since Audit		1,480.547**
		(328.235)
5 Years since Audit		1,122.686**
		(434.124)
6 Years since Audit		1,190.270*
		(515.569)
Constant	34,098.286**	34,098.639**
	(137.019)	(136.963)
R^2	.005	.005

Note. Results are from a regression of the duration since the last audit on reported taxable income. Standard errors clustered at the individual filer level are in parentheses. All regressions include individual and year fixed effects. $N = 4{,}771{,}427$.

* $P < .05$.

** $P < .01$.

lower point estimates. It should be noted that the audit outcome is not exogenous to taxpayers' characteristics. However, we believe these results are strongly suggestive of the expected behavior. In other words, while we find that noncompliant filers increase compliance after audit, we do not find that compliant filers increase noncompliance. These results are consistent with what Gemmell and Ratto (2012) observe using UK data.

Second, we interact the amount of the audit adjustment with the Postaudit dummy variable in equation (3) to see if the adjustment amount magnifies the effect of Postaudit. We report the result of this interaction analysis in Table OA4 of the Online Appendix. In Table OA4, we can see that the interaction effects are not significant when all postaudit years are grouped together. Table OA5 examines this interaction effect over time and shows that it is positive and significant in years 2 and 3 after the audit, though the magnitude of these interaction effects is also small compared with the main effect. In other words, if taxpayers receive a positive adjustment, they will increase their reported income significantly, as shown in Figure 4, but the size of the adjustment increases this effect only marginally. This should not be surprising, as the average positive adjustment in our data is $5,167 while the largest effect that audits have is an average increase in reported income after audit of around $2,850. What this means is that, on av-

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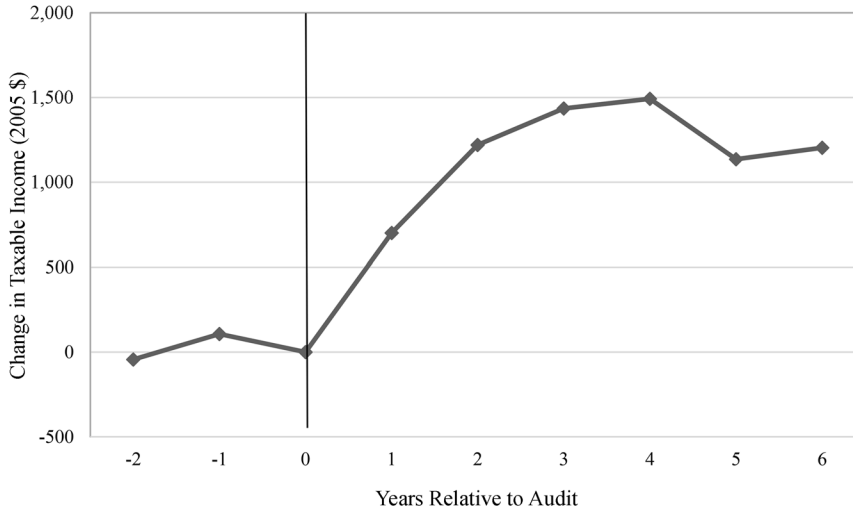


Figure 3. Effects of audit on taxable income

erage, individuals report more income following audit, but they do not increase reported income by the same magnitude as the amount of underreported income found during audit.

7. Effect under Third-Party Information and Income Volatility

We now examine whether the effects of an audit differ with the level of third-party reporting by estimating separate impacts by the source of income or the type of deduction that is being claimed. We then examine whether the response differs for taxpayers with incomes that are more or less volatile.

7.1. By Income Source

The IRS data allow us to distinguish incomes from wages and schedules C, D, and E. Among these sources, wage income information can be easily cross-checked with data reported by employers. Information on incomes from schedules C, D, and E is harder to verify, as little third-party information is available. We examine the audit effect on incomes from these sources by both intensive and extensive margins.

Intensive-Margin Effect. In Table 5 (and Figure 5), we present estimates of versions of equation (3) in which the dependent variable is income of a particular type. Note that Figure 5 presents the coefficients for the postaudit-year indicator variables in percentage terms to make the responses more comparable across income sources with very different mean values. We also restrict our sample to filers who have nonzero amounts of income of that particular type in the year of the audit, and therefore these findings can be considered intensive-margin results.

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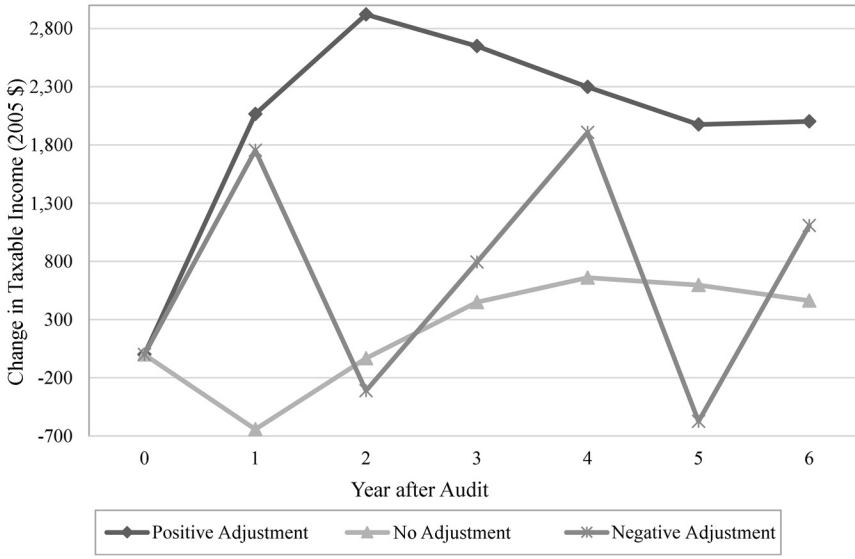


Figure 4. Effects of audit on taxable income by audit experience

In the first column of Table 5, we repeat the results from the second column 2 of Table 4, in which the dependent variable is taxable income. The results for total income are slightly smaller than, though similar to, those for taxable income.[19] This outcome is not surprising because taxable income also incorporates the filer's choice of deductions and so provides more opportunity to manipulate taxes through reporting.

In the third column, the dependent variable is wages and salaries. Similar to the results found above in the simple tabulations, this specification finds a small positive effect of an audit on reported wages, with an increase of $330–$520 in the first 3 years after audits. However, the impact on wages gradually disappears thereafter.

As noted above, schedule C (sole proprietorship) income is not generally subject to third-party reporting and so may be easier for taxpayers to manipulate. Consistent with this expectation, the estimation results in Table 5 suggest that schedule C income increases substantially after audit, by more than $1,000 in the first 2 years. Interestingly, that effect diminishes 3 years after audit and is insignificant after 4 years. Furthermore, in years 5 and 6, the estimated impact is negative. These results suggest that taxpayers with sole-proprietorship income may be more careful in reporting income immediately after audit but over time may become more aggressively noncompliant than they were in the years prior to audit.

The results for schedule D income (capital gains and losses) show no significant

[19] Total income is from form 1040, line 22, that is, adjusted gross income (AGI) with the above-the-line deductions added.

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Table 5

Effect of Auditing on Different Sources of Income

	Taxable Income	Total Income	Wage Income	Schedule C	Schedule D	Schedule E
1 Year since Audit	687.422**	606.842**	331.796+	1,172.55**	−231.878	1,011.329**
	(186.026)	(202.347)	(181.705)	(129.573)	(146.579)	(277.308)
2 Years since Audit	1,203.155**	1,036.045**	452.527*	1,035.716**	−86.169	1,498.063**
	(201.095)	(222.467)	(220.212)	(146.667)	(118.565)	(305.301)
3 Years since Audit	1,424.118**	1,280.403**	522.173*	477.465**	135.764	845.633**
	(246.485)	(277.113)	(262.200)	(154.856)	(131.805)	(313.095)
4 Years since Audit	1,480.547**	1,363.217**	103.304	4.185	284.549	356.594
	(328.235)	(367.693)	(355.860)	(196.367)	(178.504)	(417.286)
5 Years since Audit	1,122.686**	939.992+	−43.082	−723.781**	145.590	−405.062
	(434.124)	(500.293)	(505.031)	(261.934)	(212.183)	(518.825)
6 Years since Audit	1,190.27*	1,155.673+	3.137	−745.64**	133.558	−288.744
	(515.569)	(591.324)	(588.270)	(284.947)	(249.393)	(584.497)
Constant	34,098.639**	50,411.774**	41,058.233**	5,903.185**	5,729.068**	8,201.87**
	(136.963)	(152.286)	(128.911)	(72.712)	(94.025)	(129.265)
R^2	.005	.008	.009	.008	.040	.006
N	4,771,427	4,763,014	3,996,502	849,576	956,067	695,152

Note. Results are from a regression of the duration since the last audit on reported income from different sources. Standard errors clustered at the individual filer level are in parentheses. All regressions include individual and year fixed effects.

+ $P < .10$.

* $P < .05$.

** $P < .01$.

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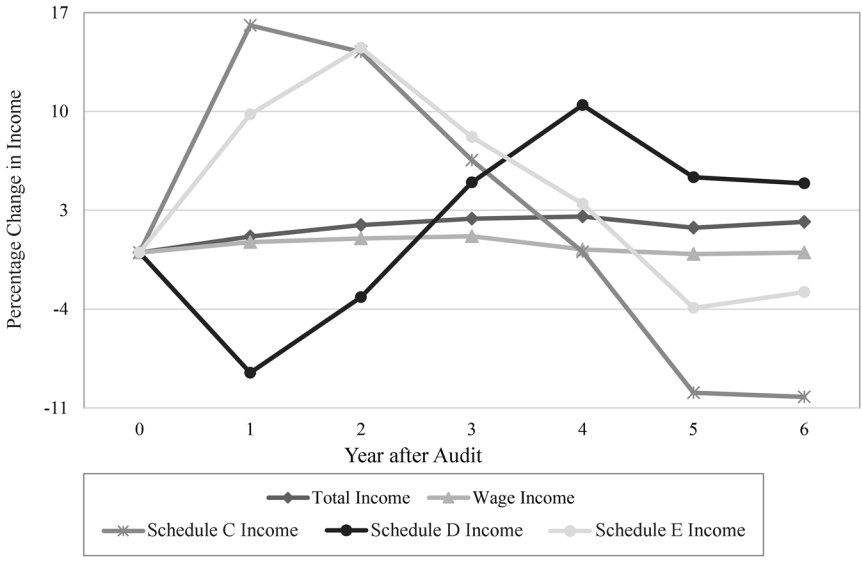


Figure 5. Effects of audit by income source

effects of audits in any year. However, the results for schedule E income, which includes partnership, S corporation, and rental income, mirror those for schedule C income. Schedule E income, like schedule C income, is largely self-reported. A significant positive impact of audits is found in the first 2 years, with the effect diminishing in the third year and no longer significant in the fourth year. In years 5 and 6, the estimates turn negative but (unlike for schedule C income) are not statistically significant in those years.[20]

Extensive-Margin Effect. To understand the extensive-margin effects of audit, we estimate linear probability models of the form

$$I\left(Y \neq 0\right)_{it} = \sum_{k=1}^{K} \beta_k (\text{Postaudit}_{it}) \times (k\text{Years since Audit}) + \gamma_i + \eta_t + \varepsilon_{it}, \quad (4)$$

where $I(Y \neq 0)_{it}$ is an indicator function equal to one if Y is not 0. The variable Y

[20] Tables OA6 and OA7 in the Online Appendix interact the Postaudit dummy variable with the amount of the audit adjustment to examine if the adjustment amount magnifies the effect of Postaudit differentially for different income sources. In Table OA6, the interaction effects are not significant except for schedule C income, which is the income source with the highest rates of noncompliance (see Table 2). The estimate in Table OA6 indicates that when the audit adjustment increases by $100, taxpayers increase their reported incomes by $1.80, which is economically small. However, two caveats apply to this estimate. First, the total estimated effect of the audit would also include the fixed amount of $609. Second, this effect is estimated on a select sample of people found to have been noncompliant on audit, and so the small effect may be reflecting that taxpayers who are more noncompliant have a greater propensity to continue to be noncompliant after audit, which leads to the small change in schedule C income. Table OA7 examines this interaction effect over time and shows that it is significant in some years for several income sources, though the magnitude of these interaction effects is also small compared with the main effect.

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represents income from schedule C, schedule E, or wages and salaries. The estimates of equation (4) are reported in Figure 6*A–D*.

The extensive-margin results for schedule C and E filers show sharp declines in the likelihood that a filer continues to file the relevant schedule after audit.[21] Thus, the increases in taxable income following audit is only one effect of audit on those with business income. The other effect is to make them less likely to claim business income.

There are at least four possible reasons for this decline, though our data do not allow us to definitively say how many filers are affected by each of the four possibilities. First, audits may result in increases in reported income and thus taxes. This effect reduces the after-tax returns on the business endeavors and may cause the filers to forgo them. Second, schedule C or E may have been filed for an activity that generated losses, which were used to offset ordinary income elsewhere on the return. An audit may have found such losses to be illegitimate, and thus the filer discontinued their use of those losses and the filing of the associated schedule.

Third, particularly for schedule C filers, an audit may have found that the filer should not have been filing as an independent contractor but instead should have been classified as an employee. The increase in compliance after an audit in this case would result in fewer schedules C being filed. Some evidence of this effect can be seen in Figure 5*C,* which shows the extensive-margin effect of audits on filers' wage income. For those who do not report wage income in the year of audit, there is a strong increase in the probability that they report wage income in subsequent years, which mirrors the results for the reporting of schedule C or schedule E income.

Fourth, for schedule C filers, an audit might induce a taxpayer to change the form of their pass-through business from a sole proprietorship to a type for which income would be reported on schedule E (such as an S corporation). Popular press reports note that schedule C income is audited at a much greater rate than schedule E income (see, for example, Bischoff 2016), and so the audited taxpayer may aim to avoid future audits by making such a change. Evidence of such an effect can be seen in Figure 5*D,* in which schedule C filers tend to be more likely to report schedule E income after audit while non-schedule-C filers are not.

Taken together, the intensive- and extensive-margin results are consistent with those of Kleven et al. (2011), who point out that taxpayers' compliance is strongly related to the ability to be noncompliant, as this ability is greatest with self-reported income such as that reported on schedules C and E.

7.2. By Deduction Item

We next estimate variants of equation (3) in which the dependent variables denote the amounts of particular types of deductions that filers claim. The first

[21] Coefficient estimates are presented in Table OA8 in the Online Appendix.

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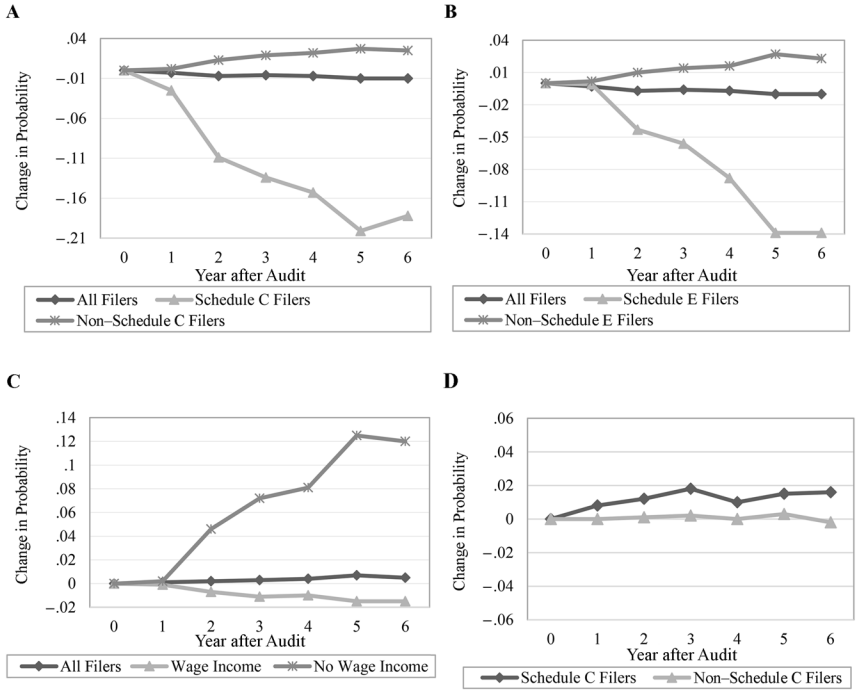


Figure 6. Effects of audit by type of filer. *A,* Likelihood of filing schedule C; *B,* likelihood of filing schedule E; *C,* likelihood of reporting wage income; *D,* likelihood of filing schedule E.

column of Table 6 presents the results for above-the-line deductions.[22] We also summarize the results in Figure 7, which plots the percentage changes in the deduction items following audit. Table 6 shows that total and itemized deductions decrease after an audit (which implies higher taxable income), with a larger decline for itemized deductions. For both types of deductions, the effects continue at a high level for up to 6 years after the audited return was filed.

As with the income-source results, the results for deductions are consistent with the theory that taxpayers manipulate their tax reporting where they are more able to do so. Thus, we see larger effects of audit on the more malleable, and less documented, itemized-deductions category. Charitable contributions are estimated to fall by up to $460 after an audit, with no decline in later years. Interestingly, although they are both subject to third-party reporting, state and local taxes and mortgage interest are also estimated to fall after an audit, with the effect increasing over time.

[22] These are deductions that are made from total income when calculating AGI and include deductions for moving expenses, health savings accounts, self-employed contributions to retirement plans and health insurance premiums, alimony paid, amounts contributed to individual retirement accounts, student loan interest, and tuition and fees paid for secondary education.

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Table 6

Effect of Auditing on Different Deduction Items

	Total Adjustments	Itemized Deductions	Charitable Contributions	State and Local Taxes	Mortgage Interest
1 Year since Audit	12.351	−562.917**	−267.326**	−39.360	82.500
	(26.780)	(119.826)	(26.915)	(28.098)	(70.714)
2 Years since Audit	−75.077*	−1,406.712**	−358.449**	−91.159**	−415.768**
	(30.054)	(128.412)	(29.488)	(32.039)	(76.328)
3 Years since Audit	−132.297**	−1,690.529**	−361.669**	−118.969**	−636.392**
	(31.941)	(139.131)	(31.393)	(37.640)	(78.949)
4 Years since Audit	−169.009**	−1,845.593**	−356.883**	−119.188*	−785.216**
	(42.228)	(172.254)	(40.350)	(52.516)	(95.295)
5 Years since Audit	−261.008**	−2,507.81**	−463.804**	−337.769**	−1,069.046**
	(57.608)	(234.789)	(55.961)	(67.109)	(127.859)
6 Years since Audit	−290.208**	−2,012.379**	−436.051**	−252.94**	−920.873**
	(58.736)	(255.428)	(60.477)	(78.813)	(132.118)
Constant	1,023.37**	13,934.544**	2,130.039**	3,026.309**	6,506.456**
	(13.565)	(57.258)	(13.775)	(17.492)	(36.246)
R^2	.031	.064	.012	.021	.076
N	1,323,803	1,931,745	1,607,515	1,859,451	1,551,268

Note. Results are from a regression of the duration since the last audit on different deductions claimed. Standard errors clustered at the individual filer level are in parentheses. All regressions include individual and year fixed effects.

* $P < .05$.

** $P < .01$.

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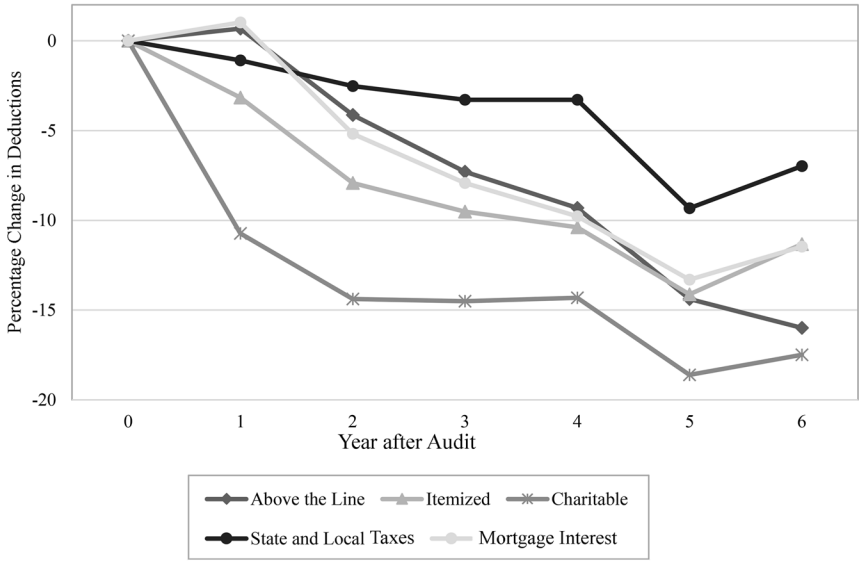


Figure 7. Effects of audit on deduction amounts by deduction type

7.3. By Degree of Volatility

The result that income sources such as schedule C and schedule E income have the largest responses to audit is not surprising given the limited amount of third-party reporting on these income sources. However, the transitory nature of the impact of audit on these income sources (and on the taxable income of those deriving income from these sources) is not as intuitive.

In addition to being subject to less third-party reporting, business income reported on schedules C and E differs from labor income (which makes up the majority of total income) by its volatility. DeBacker, Panousi, and Ramnath (2014) document that business income exhibits both a higher variance and more volatility than labor income. We believe that the difference in income processes interacts with audits in an interesting way. In particular, we posit that higher income volatility allows taxpayers to change reported income from year to year more easily. The result is that the effects of audit are not as persistent for business income as for labor income.

For example, consider an audit of a filer with only wage income. Since wage income is less volatile, that filer may believe that the IRS will be able to estimate his or her wage income for a given number of years in the future with a fair degree of accuracy. Thus, the effect of audit on wage income will be smaller and potentially more persistent. However, for the more volatile schedule C income, the filer may believe that during the same postaudit period, the IRS has very little ability

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to predict the true range of schedule C income. Therefore, the effect of audit on schedule C income is more transitory.

To test this hypothesis, we include a measure of volatility into our models and interact volatility with the audit indicator variable. We measure volatility by considering the variance in each individual filer's taxable income over time. We then group filers by filing status, number of dependents, and schedules filed. Finally, we take averages over these groups and use that calculation as the measure of volatility for each individual in that group.

Table 7 presents the results of this model for all filers and for those with schedule C income. In all cases, the interaction terms are negative, which indicates that volatility reduces the effect of audits on subsequent taxpaying behavior. This finding is consistent with our hypothesis. Furthermore, the interaction terms show that the effect of volatility tends to be more negative over time, which means that the effect of an audit is more short-lived for filers with more volatile income.

8. The Implications of Tax Sophistication

As noted above, an NRP audit is random, and the NRP subjects receive documentation that explicitly states that they were selected for a random audit used for IRS research purposes. Despite this disclosure, we find significant impacts of audit on filers. One reason for this effect may be that filers misunderstand the nature of an NRP audit. In particular, filers may believe their returns are selected for operational audits because the IRS suspects that some section of the return is incorrect, even though the letter makes clear that the IRS does not necessarily believe anyone selected for an NRP audit is delinquent in any way.[23] Alternatively, though the IRS letter is silent about whether the IRS will use noncompliance found in an NRP audit to improve the coverage of subsequent audits, some individuals may assume that the IRS will take such findings into account and adjust their behavior accordingly. To test this possibility, we look at the responses to audit across several subsamples of the tax-filing population. Each of these subsamples is created on the basis of characteristics that may be correlated with tax sophistication or understanding of tax enforcement. In particular, we proxy for tax sophistication with measures related to the number of tax returns filed. These characteristics include age and the use of a paid tax preparer. Although none of these characteristics are a perfect proxy for tax sophistication, consistent responses across these groups would suggest that tax sophistication may play a role in how filers respond to audits.

[23] While the contact letter does not make any claims about how the NRP audit will affect future audit rates, we find that the likelihood of future audit is very similar for those in the control group and the treatment group. We also see little difference in future audit rates whether the NRP found positive or negative adjustments.

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Table 7
Effect of Audit and Income Volatility on Subsequent Taxable Income

	All Filers		Schedule C Filers	
	(1)	(2)	(3)	(4)
1 Year since Audit	687.422**	1,938.009**	1,221.978**	3,492.192**
	(186.026)	(267.303)	(418.918)	(572.575)
2 Years since Audit	1,203.155**	2,865.275**	2,860.921**	5,781.773**
	(201.095)	(289.397)	(468.607)	(630.918)
3 Years since Audit	1,424.118**	3,875.297**	2,639.729**	7,303.227**
	(246.485)	(344.742)	(526.023)	(708.186)
4 Years since Audit	1,480.547**	2,743.992**	3,117.531**	6,558.445**
	(328.235)	(426.973)	(715.627)	(894.466)
5 Years since Audit	1,122.686**	3,671.93**	2,614.918**	8,842.014**
	(434.124)	(554.753)	(993.198)	(1,374.814)
6 Years since Audit	1,190.27*	2,439.411**	2,120.874*	6,669.378**
	(515.569)	(645.825)	(1,052.582)	(1,355.481)
1 Year since Audit × Volatility		−440.566**		−523.944**
		(94.452)		(163.790)
2 Years since Audit × Volatility		−580.613**		−667.382**
		(100.096)		(173.935)
3 Years since Audit × Volatility		−857.054**		−1,063.773**
		(106.480)		(184.120)
4 Years since Audit × Volatility		−426.451**		−764.784**
		(136.857)		(230.788)
5 Years since Audit × Volatility		−866.402**		−1,395.788**
		(179.025)		(323.981)
6 Years since Audit × Volatility		−415.373*		−1,003.972**
		(197.789)		(321.598)
Constant	34,098.639**	34,065.13**	43,563.92**	43,524.824**
	(136.963)	(135.957)	(281.317)	(279.661)
R^2	.005	.006	.003	.004
N	4,771,427	4,771,427	849,576	849,576

Note. Results show the effect of the interaction of the duration since the last audit and income volatility on reported income. Columns 2 and 4 include average volatility of the demographic group of the tax filers. Standard errors clustered at the individual filer level are in parentheses. All regressions include individual and year fixed effects.

* $P < .05$.

** $P < .01$.

8.1. Responses by Filers' Age

Figure 8 splits the sample according to the age of the primary filer in the year of the audit.[24] Here, the three oldest age groups (35–44, 45–54, and 55–64) appear to be the most responsive after an audit, with estimated effects 2 years after audit in excess of $1,200, while the impact for the youngest group 2–4 years after audit is approximately $750. These results contrast with those of Kleven et al. (2011), who find that the propensity to underreport income falls with age.

[24] For observations in the control group, we split according to age in the tax year for which they were drawn as a control observation. Table OA9 in the Online Appendix shows the regression results from which Figure 8 was created and also shows the significance of these postaudit effects.

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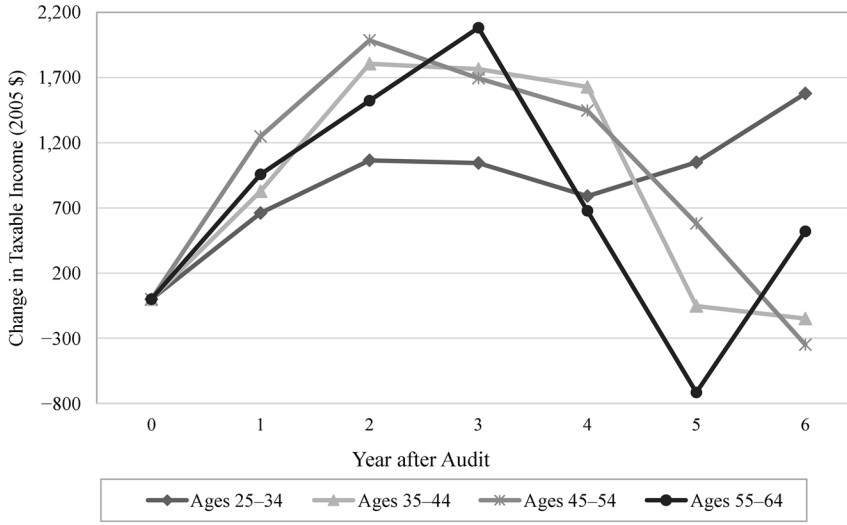


Figure 8. Effects of audit on taxable income by filer's age

Considering the long-term patterns, we see that the youngest age group (25–34) responds significantly differently than the older age groups to audit. In particular, this age group does not show a return to the preaudit trend in income reporting. Following an audit, those ages 25–34 report higher incomes, and this effect persists over the next 6 years with increases generally over the entire window. In contrast, the older age groups show an increase in reported taxable income in years 2 and 3 after the audit but the effects generally decline afterward. Thus, we find evidence that audits have more persistent deterrence effects on those who are relatively new tax filers. This finding may reflect differences in how those with a short filing history update their prior beliefs about IRS enforcement following an NRP audit as compared with those with longer filing histories and more experience with IRS enforcement.

8.2. Responses by Use of Paid Preparer

We now condition samples on whether the taxpayer used a paid preparer to file the tax return. Our assumption is that those using a paid preparer have more information about the tax system. The paid preparer is an expert and someone who likely has participated in the filing of a large number of tax returns. Thus, a taxpayer employing the help of a paid preparer will be a filer who is more tax literate since she has the assistance of the preparer.

Figure 9 displays the results comparing those who used paid preparers in the year the audited return was filed with those who did not.[25] In the first 3 years af-

[25] Table OA10 in the Online Appendix shows the regression results from which Figure 9 was created and also shows the significance of these postaudit effects.

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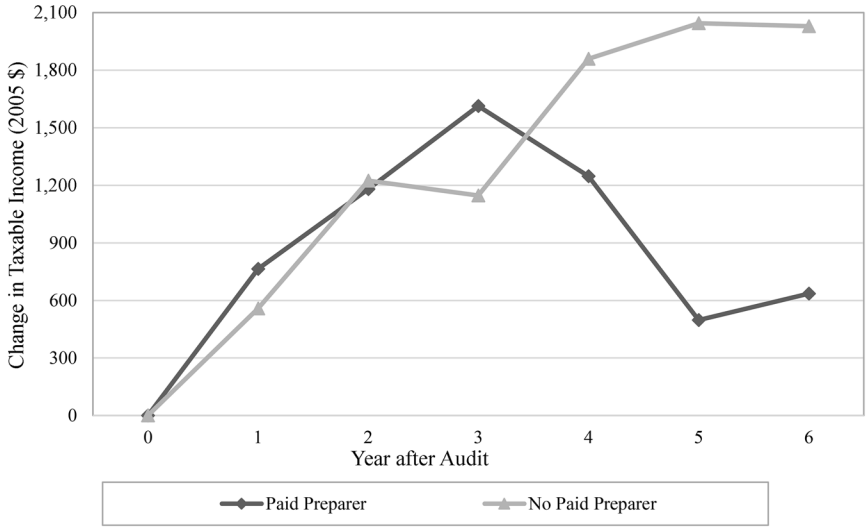


Figure 9. Effects of audit by use of paid tax preparer

ter audit, these two groups have very similar responses, increasing reported taxable income by more than $1,000 above that of the comparable group that was not audited. However, after year 3, the postaudit trends diverge. In these later years, the trend for those using a paid preparer moves taxable income toward its preaudit level, while for those who did not use a paid preparer, the increase in reported taxable income is persistent. That is, the increase in reported income is much more transitory for taxpayers using a paid preparer. This is consistent with the theoretical prediction of a larger response from those with less experience with the tax system. In particular, it is likely that paid preparers better understand the random nature of the NRP audits and thus initially have their clients increase reported income (for example, during the time the audits might still be underway) but then return to preaudit reporting behavior knowing that the NRP audit signals nothing about future audit probabilities.

8.3. Responses by Income

Finally, we might expect that taxpayers' sophistication may vary with income. Taxpayers with higher incomes may rationally invest in a deeper understanding of the tax code and may therefore have a differential response to audit. To understand these differences, we create treatment and control groups according to the quintile of their income immediately prior to audit.[26]

[26] A related topic of interest would be how audit response depends on marginal tax rate, since the predictions of the Allingham and Sandmo (1972) model are sensitive to details of the tax code. However, this introduces endogeneity concerns that would potentially swamp the usefulness of our randomized treatment. To that end, we point toward the positive relationship between income tax

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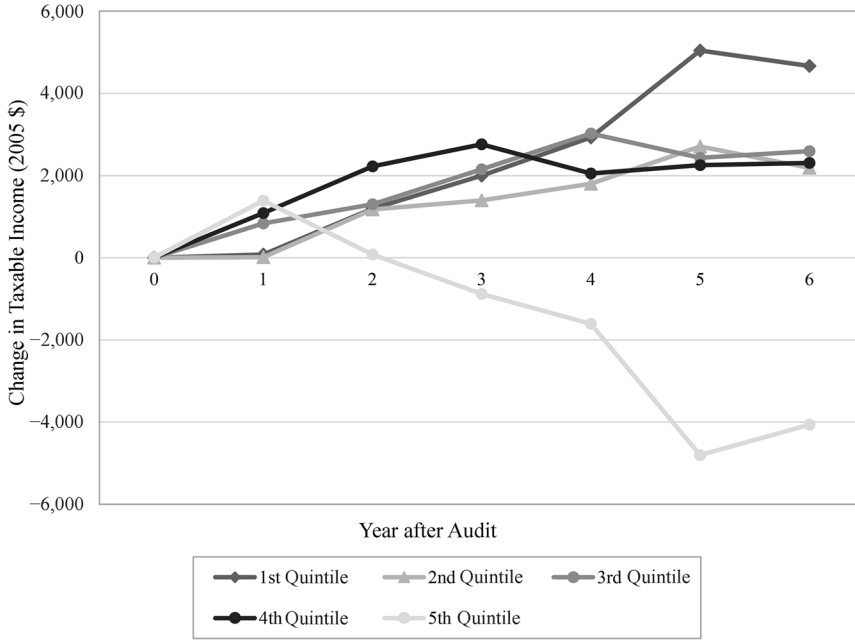


Figure 10. Effects of audit by income quintile

Figure 10 shows how the responses to audit differ across income groups. Income quintiles 1–4 show similar responses to audit, with persistent increases in reported taxable income following audit. Among these, the lowest income group has the strongest response to audit measured in the difference in dollars of taxable income reported after audit. Note that this is true despite this group having the lowest taxable income and marginal tax rates at the time of audit. Most fascinatingly, the top income quintile shows the opposite trend, with a persistent decline in reported income after the initial increase in income in the year after the audited return was filed. This response of the highest income group is similar to results in Slemrod, Blumenthal, and Christian (2001), which finds that lower income taxpayers increased their tax payments when informed by the tax authority that their returns would be "closely examined," while high-income taxpayers decreased their tax payments. The high-income response is also similar to the response by corporations found in DeBacker et al (2015). High-income individuals appear to be more likely to know, then, that these are random audits whose

rates and income and claim that results based on income quantiles are probably quite similar to results based on tax rates.

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results are not used to determine whether they are audited in the future, and so they do not increase their reported income in response.[27]

9. Conclusion

Tax evasion is estimated to lower global tax collections by about 18 percent and result in approximately $3 trillion in lost public funding worldwide (Murphy 2011). Among all countries, the United States suffers the largest loss in absolute terms because of the sheer size of its economy. A key measure to confront this problem is through audits, which have two main effects on the audited. First, there is an immediate revenue gain when the auditors discover noncompliance. Second, those audited tend to report higher taxable income in subsequent years, which results in further revenue gains. Our study rigorously evaluates these effects and identifies several factors determining the effectiveness of tax audits in the short and long term.

To do so, we examine the randomized audits by the IRS's NRP during the 2006–9 period. Our simple difference-in-differences specification indicates that an audit increases reported taxable income by more than $1,100 per year, equivalent to 2.9 percent of the average income. This effect is only 1.3 percent for wage income but is 13 percent for schedule C income. Furthermore, we find that the impact of auditing on reported wages lasts over time but is fleeting for schedule C income.

Similar results are found when controlling for individual fixed effects. These results suggest that AGI increases for at least 6 years after an audit. Contributing to this increase, schedule C and schedule E income (which is not subject to third-party reporting or withholding) tends to sharply increase after an audit, but this increase diminishes (and turns negative) 5 or more years after audit, while the increase in wages and salary (which are subject to third-party reporting and withholding) is considerably smaller. In addition, above-the-line and itemized deductions both decrease significantly after audit, and the decrease in deductions is apparent even among deductions (for example, state and local taxes and mortgage interest) that are subject to third-party reporting.

It is worth noting that the effects we measure are the effects of random audit. In theory, filers may respond differently to nonrandom audit; for instance, they can make some inferences about the auditing process from the fact that they were selected. However, randomizing audits provides us with a much more reliable setting to estimate the impact. One argument that suggests the effects of nonrandom and random audits may not be far apart is made by Manoli and Turner

[27] Slemrod, Blumenthal, and Christian (2001) speculate that their perverse findings for high-income taxpayers may be due to high-income taxpayers believing that a higher reported income will lead to a lower probability of audit, and so if they know that they are going to be audited, it no longer makes sense to increase their reported income. Alternatively, they note that the taxpayer may view the audit as a negotiation, and so it makes sense to start with a low bid. Although the first mechanism is unlikely to explain our results, the second mechanism might, in that after going through an NRP audit, high-income taxpayers might believe that they would pay less tax after audit if they reported less income on their returns.

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(2014). They provide evidence from a randomized field experiment showing that the content of contact between the IRS and tax filers is much less important than the existence of this contact.

There are several clear implications one can draw from the results of this study. First, an audit of a randomly selected individual tax filer increases reported taxable income by roughly 2.9 percent, and this effect appears to persist for at least 6 years. Second, audits produce more transitory effects on income sources that are subject to less third-party reporting. Third, income volatility provides an important shield for those willing to evade taxes. Fourth, responses to audits are affected by the tax sophistication of the filer. These results help to better explain the information problem in law enforcement and identify what types of intervention may be most effective.

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